Chapter 10

Propagation, Purification, and In Vivo Testing of Oncolytic Vesicular Stomatitis Virus Strains

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Abstract

Oncolytic viruses are self-amplifying therapeutics that specifically replicate in and kill cancer cells. We have previously shown that vesicular stomatitis virus (VSV) can be used as an oncolytic virus. A strain of VSV harboring a mutation in the M protein (VSVΔ51) was found to exhibit enhanced tumor selectivity over its wild-type counterpart due to its inability to overcome antiviral programs in normal cells and due to the frequent defects in antiviral signaling pathways observed in the majority of tumors. VSVΔ51 can harbor transgenes, is easily propagated and purified to high titers, and shows potent oncolytic activity in several mouse models, including syngeneic CT26-lacZ subcutaneous colon carcinoma models. However, VSV-neutralizing antibodies targeting mainly the VSV-G surface glycoprotein arise within 3–5 days following the initial dose. This should be considered for strategies aiming at increasing the effectiveness of VSV through delivery of additional doses of virus or aiming to prolong VSV replication in vivo.

Key words: Vesicular stomatits virus, Oncolytic, Purification, Syngeneic mouse model, Neutralizing antibodies

1. Introduction

Vesicular stomatitis virus (VSV) is a small negative-strand RNA virus belonging to the Rhabdovirus family. The VSV genome (approximately 11 kbp) encodes five distinct genes referred to as the nucleocapsid (N), polymerase proteins (P and L), the matrix protein (M), and the surface glycoprotein (G). The VSV virion is bullet-shaped and consists of viral RNA wrapped tightly within the ribonucleocapsid (an array of N, P, L proteins) to which binds M protein. The virion becomes enveloped with G protein trimers

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upon budding from the host cell, which is the main target of neutralizing antibodies generated against the virus (1).

VSV has been explored as an oncolytic virus due in part to its rapid life cycle and its potent cytolytic activity. Being an RNA virus that replicates in the cytoplasm, VSV is unlikely to recombine with cellular DNA and cannot integrate the genome. Combined with low human exposure to the virus and the absence of preexisting immunity in the population, this makes VSV an attractive candidate for human cancer therapy (2).

Interferon (IFN) signaling is thought to be defective in approximately 70–75% of all tumors (3). In light of this, we and others have explored VSV strains that exploit this tumor-specific defect to enhance tumor targeting (4, 5). One of these mutant strains is VSVΔ51 which harbors a mutation in the M protein at methionine 51. Whereas wild-type M protein can block the nuclear export of cellular antiviral genes and their subsequent expression, the MΔ51 protein cannot and so VSVΔ51 replication is quickly subdued in normal cells that have intact antiviral signaling pathways (4).

2. Materials

2.1. Cell Culture and Virus Propagation

1. HyQ High glucose Dulbecco's modified Eagle medium (DMEM) supplemented with 10% fetal bovine serum (FBS) (HyClone, Waltham, MA).
2. Solution of trypsin (0.25%) and ethylenediamine tetraacetic acid (EDTA) (1 mM) from Gibco/BRL.
3. ViCell (Beckman/Coulter) automated cell counter (alternately any standard hemocytometer can be used).
4. Phosphate buffer saline (PBS, HyClone).
5. 0.20 μm Bottle top filtration unit (Millipore).

2.2. Virus Purification

1. OptiPrep 60% (w/v) (Sigma).
2. Tris stock solution (100 ml): Dissolve 12.1 g Tris base into 100 ml ddH_2O. Filter and store at 4°C.
3. EDTA stock solution (100 ml): Dissolve 3.72 g EDTA $Na_2{\cdot}2H_2O$ into 100 ml ddH_2O. Filter and store at 4°C.
4. NaCl stock solution (100 ml): Dissolve 5.84 g NaCl into 100 ml ddH_2O. Filter and store at 4°C.
5. Solution B (100 ml): To 50 ml water, add 30 ml Tris stock solution and 3.0 ml EDTA stock solution. Adjust pH to 7.4 with 5 M HCl, and adjust volume to 100 ml with ddH_2O. Filter and store at 4°C.

6. Solution C (100 ml): To 50 ml water, add 10 ml NaCl stock solution, 5.0 ml Tris stock solution, and 0.5 ml EDTA stock solution. Adjust pH to 7.4 with 1 M HCl, and adjust volume to 100 ml with ddH_2O. Filter and store at 4°C.
7. Solution D: Add 5 volumes of OptiPrep (Sigma) to 1 volume solution B.
8. Gradient maker (e.g., Auto Densi-Flow, LabConco).
9. Ultra-clear thinwall Beckman ultra-centrifuge tubes (11-ml maximum volume) for use with Beckman SW41Ti rotor.

2.3. Virus Titering

1. 2× concentrated DMEM prepared from powder (Gibco/BRL) supplemented with 20% FBS.
2. 1% agarose dissolved in deionized water.
3. 3:1 methanol: Glacial acetic acid fixative solution.
4. Coomassie brilliant blue solution: 0.5 g Coomassie Brillant Blue R (Sigma) is dissolved in 250 ml 40% methanol and subsequently filtered on Whatman paper. 250 ml 20% acetic acid is then added to the filtered solution.

2.4. Mouse Tumor Implantation, Treatment with Virus, and Imaging

1. Syringes.
2. IVIS200 Series Imaging System (Xenogen).
3. 3% Isoflurane (Baxter).
4. D-Luciferin, 10 mg/ml in PBS (Molecular Imaging Products).
5. Electronic caliper.

2.5. Collection of Mouse Serum and Neutralizing Antibody Assay

1. Restraint tube (perforate 50-ml Falcon tube).
2. Vaseline.
3. Beveled syringe needle.
4. Heparinized collection tubes.

3. Methods

A key consideration in using VSV or VSVΔ51 for both in vitro and in vivo experimentation is starting from a pure high-titer stock of virus containing as little defective uninfectious particles as possible. To this end, virus is propagated by infecting sensitive cells (e.g., Vero cells) at a low multiplicity of infection and then collected from supernatants prior to extensive cell lysis. The virus is subsequently purified on gradient to eliminate cellular contaminants, such as DNA, protein, and cellular exosomes (6, 7), which can induce antiviral programs and impact the permissiveness of cells in vitro or induce immune responses in vivo. It is possible to quantify how many infectious VSV particles have been obtained using viral plaque assays. Once viral titers have been assessed, virus can then be used for in vitro or in vivo testing alone or in combination with other agents (8, 9). In vitro, virus spread can be followed easily when VSV is engineered to express fluorescent proteins using fluorescence microscopy. For example, we use VSV variants in which have been inserted green fluorescent protein (GFP) or firefly luciferase (FLuc) between the G and L genes driven by an intergenic VSV promoter. FLuc-expressing variants are useful as they can be imaged in vivo in real time using injectable luciferin and in vivo imaging systems (IVISs).

The oncolytic properties of VSV can be observed in xenografts implanted in immunocompromised animals or in immunocompetent mice harboring syngeneic tumors. The latter is preferred since immune responses can both positively and negatively impact oncolytic virus efficacy (10, 11). Subcutaneous models are convenient since tumors can be injected directly and volumes post infection can be followed using a caliper (Fig. 3). Viral replication can be followed by IVIS as well. Infection of mice with VSV leads to production of neutralizing antibodies from B cells that typically recognize the G-protein. These antibodies effectively inactivate the virus and generally arise 4–5 days post infection when virus is given intravenously (Fig. 4). VSV-neutralizing antibody titers can be easily measured from sera extracted from mouse blood obtained by saphenous vein bleeds at various times post infection.

3.1. Propagation of VSV in Cell Culture

1. Prior to infection with VSV, Vero African green monkey kidney cells are seeded at 1×10^4 cells per cm^2 in FBS supplemented with DMEM in 150-mm cell culture petri dishes (25 ml per plate).
2. When cultures have reached roughly 95% confluency (~3–4 days), they can be infected with a preexisting virus stock at 2×10^5 plaque forming units (PFUs) per 150-mm petri (an MOI of ~0.01, see Note 1).
3. For infection of ten plates (see Note 2), 2×10^6 PFUs of stock is diluted in 50 ml of serum-free DMEM and vortexed thoroughly (see Note 3).
4. Supernatant is removed from each petri dish and 5 ml of virus mixed in serum-free DMEM is added to each plate. Petris are then placed in a 37°C 5% CO_2 humidified incubator for 45 min (see Note 4).
5. After incubation, the serum-free media is removed and 25 ml of DMEM containing 10% FBS is added to each petri. Infected cultures are then placed in a 37°C 5% CO_2 humidified incubator.

6. When approximately 50% cytopathic effect (CPE) (cell rounding) is observed (usually, after approximately 24 h), the supernatants are collected in 50-ml Falcon tubes (see Note 5).
7. The supernatants are centrifuged at 780 × *g* for 5 min to pellet heavy cellular debris.
8. Cleared supernatants are filtered through 0.2 μm membranes (Millipore) and then either used directly or purified further (see Note 6).

3.2. Purification of VSV

Purified VSV crude obtained in Subheading 3.1 should be further concentrated by ultracentrifugation prior to gradient purification.

1. Virus can be pelleted using a Beckman Coulter Avanti J25I Ultracentrifuge at 14,000 rpm (28,000 × *g*) for 1 h 30 min at 4°C in polypropylene centrifuge bottles (JA-14 rotor). The virus pellet should be visible at this stage and supernatant should be carefully discarded, removing as much media as possible without disturbing the pellet (see Note 7).
2. To purify the virus using OptiPrep® (see Note 8), the virus pellet isolated following ultracentrifugation is resuspended in maximum 2 ml of solution C (see Note 9).
3. For a single 12-ml gradient (accommodating up to 2 ml of resuspended virus), prepare 5 ml of 15% OptiPrep® solution by diluting 1.5 ml of solution D (which is 50% OptiPrep®) with 3.5 ml of solution C. Also prepare 5 ml of 35% OptiPrep® solution by diluting 3.5 ml of solution D with 1.5 ml of solution C.
4. Transfer 5 ml of the 35% OptiPrep® solution into the first chamber of a gradient maker (the chamber closest to the exit valve on a LabConco Auto Densi-Flow model gradient maker) and 4.5 ml of the 15% OptiPrep® solution into the second chamber (furthest away from the exit valve).
5. Let ~0.5 ml of the 35% solution leave the exit valve before opening the valve connecting the two gradient chambers. This generates 9.5 ml 15–35% continuous OptiPrep® gradient which can be collected in clear SW-41 Ti thin-walled ultracentrifuge tubes (Beckman # 344059).
6. When the gradient has been obtained, carefully layer up to 2 ml of virus suspension onto the gradient.
7. Ultracentrifuge virus and gradient at 36,000 rpm (160,000 × *g*) for 1.5 h at 4°C.
8. If the pellet has been properly dispersed, a single band should be visible roughly mid-gradient (see Fig. 1). Collect this band by first removing the OptiPrep® above it and then aspirating the band carefully using a P 1000 pipet (should take at least 600 μl).

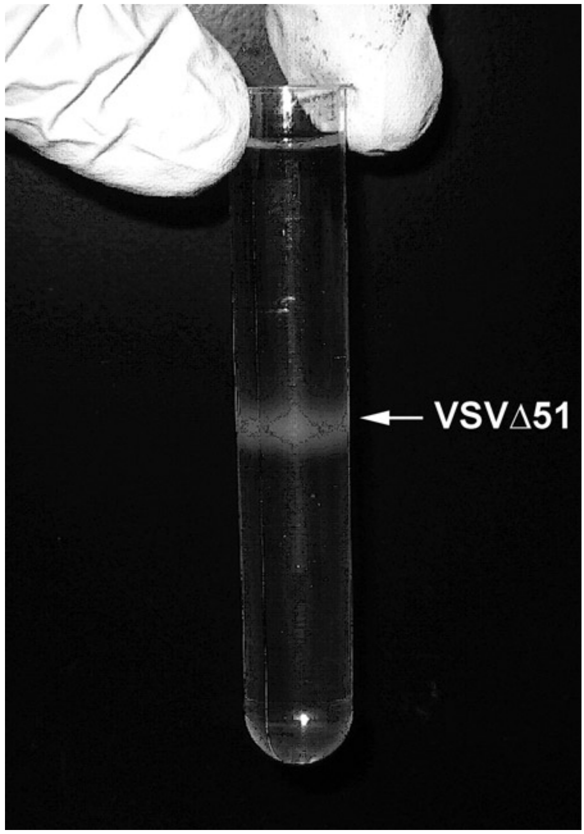

Fig. 1. VSV purification on OptiPrep® Gradient. VSV band in a 5–50% OptiPrep Gradient obtained starting from 10 × 150-mm petris of Vero cells following a 24-h incubation. Note that a single band is obtained approximately mid-gradient. Multiple bands, some with macroscopic flakes, may be observed if pellet is insufficiently dispersed in solution C prior to gradient purification.

9. To avoid repeated freeze-thawing, make many small-volume aliquots (e.g., 10 μl) and store at least at −80°C. This virus should remain stable for several months.

3.3. Quantification of Infectious VSV Particles by Viral Plaque Assay

VSV obtained from various sources, including infected cell cultures, tissues, and purified virus stocks, can be quantified by viral plaque assays.

1. Vero cells are plated in six-well plates at densities such that they have reached approximately 95% confluence the following day. In six-well plates, this equates to approximately 1×10^6 cells per well (see Note 10).
2. Serum-free media is then used to do serial dilutions of virus (see Note 11).

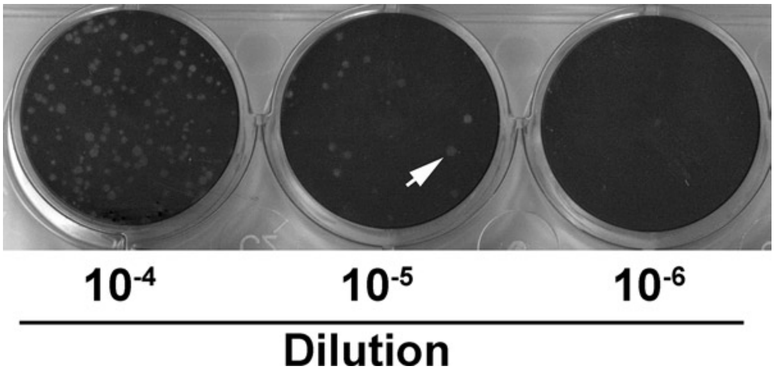

Fig. 2. Viral plaque assay for quantifying VSV titers. Vero cells were plated and inoculated with serial dilutions of a supernatant obtained from VSVΔ51-infected cells then coated with an agarose/DMEM overlay. Twenty-four hours later, wells were fixed with methanol–acetic acid fixative, agarose overlay was removed, and fixed cells were stained with Coomassie brilliant blue. The *arrow* points to a typical plaque. The middle well contains 25 plaques, an appropriate number to calculate viral titers which in this case comes to $(25/10^{-5}) \times 2 = 5 \times 10^{6}$.

3. The media covering the plated Vero cells is then removed and 500 μl of diluted virus stock (for each dilution) is used to infect Vero cells which are then incubated for 45 min at 37°C in a 5% CO_2 humidified incubator.
4. During this time, a 1% agarose solution prepared in sterile-deionized H_2O should be brought to a boil and put in a 42°C water bath. 2× concentrated DMEM containing 20% FBS should be warmed in a 37°C water bath as well.
5. After the 45-min incubation, the media covering infected Vero cells should be removed. 1:1 volumes of 1% agarose: 2× DMEM, 20% FBS at 42°C should then be mixed together and 1–2 ml of this mixture should be used to cover each well of Vero cells infected with virus dilutions (see Note 12).
6. Cells should then be put back in a 37°C 5% CO_2 humidified incubator for another 24 h (see Note 13).
7. After incubation, 1 ml methanol–acetic acid fixative (3:1 ratio) is added on top of the agarose layer in each well and allowed to incubate at room temperature for 1 h.
8. Subsequently, the agarose can be washed or lifted off the wells using a strong water current (e.g., under a tap).
9. Fixed Vero cells are stained using 1 ml of a Coomassie blue solution and incubated for 30 min at room temperature on a plate shaker (at low speed).
10. Resulting viral plaques can be easily visualized (see Fig. 2) and plates can be stored indefinitely.
11. To calculate viral titers, plaques are counted ideally at a dilution step, where between 10 and 100 plaques are visible.

12. The number of plaques is then divided by the appropriate dilution and the resulting number is multiplied by 2 to give a titer in PFU/ml (since 500 μl were used to infect cells). For example, if 25 plaques were counted in the well where a $1/10^6$ dilution was used, the titer of the initial undiluted sample is $(25/(10^{-6}))\times 2 = 5\times 10^7$ PFU/ml.

3.4. In Vivo Testing of Oncolytic VSVΔ51

VSVΔ51 has been shown to be an effective oncolytic in several tumor models. Because it is well-recognized that immune responses play a major role in oncolytic virus efficacy, it is best to study this oncolytic strain in immunocompetent models. One model we frequently use is the CT26 colon carcinoma model, which is syngeneic in Balb/C mice (see Note 14). This model is convenient because tumors grow homogenously with relatively good reproducibility without ulcerating or metastasizing too quickly. As such, they are particularly suitable for measuring tumor volumes.

1. Balb/C mice can be injected subcutaneously in the flanks with 3×10^5 cells (>95% viable) resuspended in 100 μl serum-free media or PBS.
2. Ten days later, tumors reach approximately 200 mm^3 as measured using a caliper and the formula Volume = $(\text{Length}\times(\text{Width})^2)/2$.
3. VSVΔ51 can be injected either intratumorally or intravenously at doses of up to 1×10^9 PFU (see Note 15).
4. Tumor volumes may be measured at regular intervals. Delayed tumor progression can be observed even after a single injection of VSVΔ51 delivered intratumorally (Fig. 3).
5. Virus replication can be followed in vivo using IVIS if VSVΔ51-expressing Fluc is used. To do this, mice are injected with 200 μl of luciferin substrate (10 mg/ml) intraperitoneally. Five minutes later, mice are anesthetized using isoflurane and then put in the IVIS for photon measurement (see Note 16). Exposure times vary generally between 30 s and 3 min.
6. If so desired, tumors or other organs (for toxicity studies, for example) can be collected by dissection and used as a starting material for viral plaque assays to confirm the presence (or absence) of virus.
7. To do this, tissues (whole or partial) are dissected, weighed, and homogenized in 500 μl PBS. It is important to drench mice in ethanol 70% prior to dissection to minimize the risk of contamination (see Note 17).
8. A fraction of this homogenate can be used for viral plaque assays (Subheading 3.3, see Note 18).

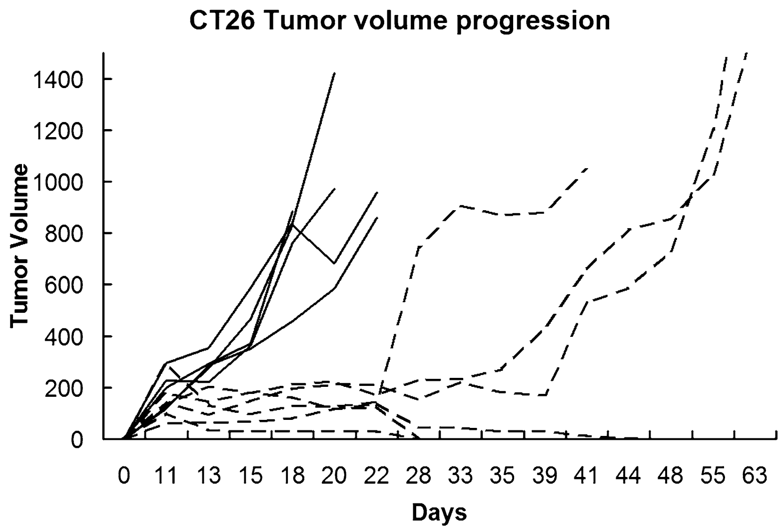

Fig. 3. Injection of VSVΔ51 intratumorally delays progression of syngeneic CT26 tumors. Mice were implanted with 3×10^5 VSVΔ51-sensitive CT26-lacZ cells (see Note 14). Tumors were injected with 5×10^7 PFU of VSVΔ51 (in 50 μl PBS) 11 days following tumor implantation. Tumor volumes were measured every 2–3 days using an electronic caliper. Volumes are expressed in mm^3 and were calculated according to the formula in Subheading 3.4, step 2. Five and six mice were included in PBS (*full lines*) and VSVΔ51 (*dashed lines*) groups, respectively.

3.5. Following the Production of VSV-Neutralizing Antibodies In Vivo

VSV is quite sensitive to neutralizing antibodies present in the serum of immunized individuals; as such, additional doses of VSV may not reach tumors following the rise of neutralizing antibody titers. It may be useful for investigators to monitor neutralizing antibody titers, for example in response to cotreatment with drugs that can impact innate and elicited immune responses, such as cyclophosphamide (12). To measure VSV-neutralizing antibodies, fresh whole blood from infected animals or individuals is required. In mice, whole blood can be collected repeatedly by saphenous bleed.

1. Mice are put facing downward in a restraint tube (e.g., perforated 50-ml Falcon tube).
2. The mouse tail is wrapped around the index finger and held by the middle finger.
3. Light pressure is applied by the index finger and the middle finger on the lower abdomen and legs toward the wall of the restraint tube, securing the mouse in place. At this stage, the mouse should have its head in the restraint tube and its hind legs sticking out of the tube.
4. Using the thumb, light pressure is applied just above the hip, extending the leg.
5. Vaseline is then applied to the extended leg so that the saphenous vein can be easily visualized.

6. A small puncture at a 45° angle is done directly on the saphenous vein and blood is drawn using heparinized collection tubes (see Note 19).
7. Blood contained within the capped heparinized collection tubes is spun down at 3,400 × *g* in an Eppendorf minifuge for 5 min to separate serum from blood cells.
8. Serum (fraction on top) should be transferred gently to a fresh tube using a micropipet (see Note 20).
9. Collected serum can be tested for its ability to neutralize VSV infection (see Note 21).
10. For the neutralizing antibody assay, Vero cells are plated in 96-well plates at a density of 12,500 cells per well in 100 μl of FBS supplemented with DMEM and incubated overnight at 37°C in a 5% CO_2 humidified incubator.
11. The following day, each serum sample is diluted 1/50 in serum-free medium (see Note 22).
12. 100 μl of the diluted serum preparation is plated into an empty 96-well microtiter plate.
13. In each subsequent row, 50 μl of serum-free DMEM is plated.
14. 50 μl from the 1/50 dilution is transferred and serially diluted (1:2) across the plate. 50 μl is discarded from the last row.
15. Subsequently, 2×10^5 PFU VSV per 50 μl serum-free DMEM is added to each well (see Note 23). A negative control, where virus-free DMEM is used, and a positive control, where no serum is put in the initial 1/50 dilution, should also be included. A schematic representation of a typical dilution setup is shown in Fig. 4a.
16. The mixtures of serum dilutions and virus are incubated for 1 h at 37°C.
17. Media is removed from 96-well plates containing Vero cells, and the contents of the 96-well plate containing virus:serum mixtures are immediately transferred.
18. Cells and virus:serum mixtures are incubated at 37°C in a 5% CO_2 humidified incubator for 1 h.
19. 100 μl of DMEM containing FBS is then added to all of the wells, and the plates are further incubated for 48 h.
20. After this time, CPEs can be visualized or quantified by a variety of methods, including visual inspection (cell rounding), Coomassie blue stain, or using metabolic dyes, such as MTT or Alamar Blue®.
21. Antibody titers in a given serum sample can be determined by assessing the dilution at which 50% CPE is observed. Figure 4b shows the progression of neutralizing antibody titers collected form Balb/C mice injected intravenously with 1×10^7 PFU of VSVΔ51. In this case, the same virus was used for the neutralization antibody assay.

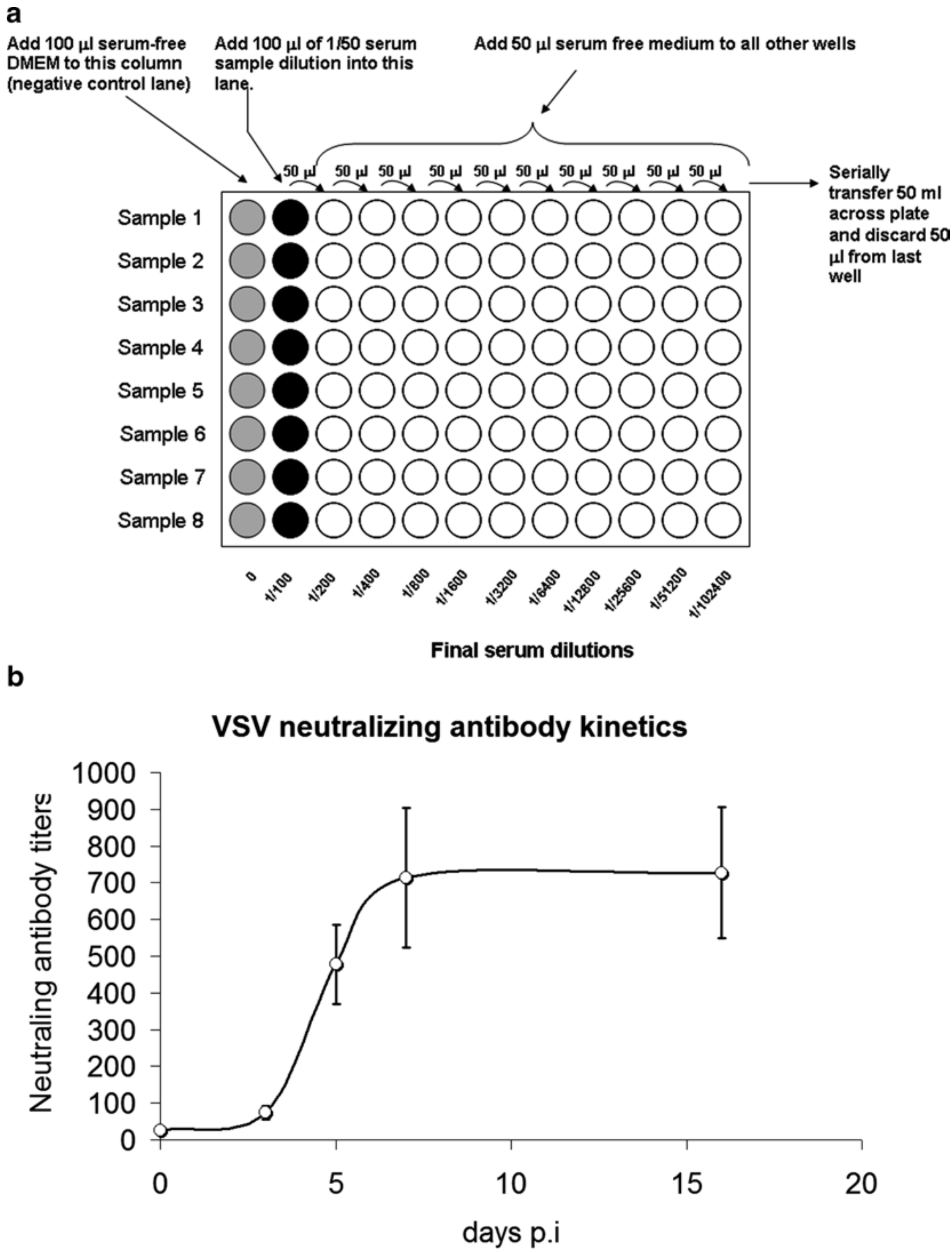

Fig. 4. VSV-neutralizing antibody titers following intravenous administration of VSVΔ51 in mice. (**a**) Typical experimental setup for determining neutralizing antibody titers. (**b**) Balb/C mice were injected with 1×10^7 PFU of VSVΔ51 (in 100 µl PBS). Serum was collected over time by saphenous vein bleed, and VSV-neutralizing antibody titers were determined using the method described in Subheading 3.5. In this case, Alamar Blue® metabolic dye was used to assess Vero cell viability. Five mice were included in each group and *error bars* represent standard error. Note that in this experiment, neutralizing antibody titers remained at their high plateau for at least 56 days post infection.

4. Notes

1. In general, high-titer seed stocks tend to yield more viral progeny.
2. Typically, we use 10×150-mm petris which are sufficient for routine use, although more stock may be necessary for in vivo experiments.
3. Brief vortexing of stocks prior to dilution is always recommended as concentrated viruses may deposit or clump. Vortexing after each dilution is also recommended.
4. Low-volume infection is necessary in order to optimize virus adsorption. During this step, it is important that cells do not dry and so the best results are obtained when plates are shaken lightly every 5 min.
5. Longer incubation times give higher yields, but also increase the number of defective particles and potential cellular contaminants.
6. The filtrate contains VSV crude that can be titered (see Subheading 3.3) and used in in vitro experiments if so desired. It is nonetheless recommended to further purify and concentrate the virus by ultracentrifugation and better yet to purify the virus on gradient.
7. At this stage, if so desired, the virus pellet can be resuspended in 1 ml PBS solution and frozen at −80°C for use at a later time. This purified crude can be titered and used directly in vitro or in vivo. Because some contaminants can remain in purified crude, best results are obtained following purification on either sucrose or optiprep gradient.
8. Because sucrose can be toxic when injected in mice, we use OptiPrep® (Iodixinol) in order to be able to use the same virus stock for in vitro and in vivo experiments. Purification on optiprep gradient yields the highest titers when initiated from fresh purified crude (as opposed to frozen) since each freeze–thaw cycle of virus stocks leads to approximately two- to five-fold decreases in titers.
9. Obtaining a homogenous virus preparation without clumps is critical at this stage since clumps may lead to multiple bands to extract in the gradient. To this end, it is recommended to allow the resuspended pellet to disperse overnight at 4°C. Also, rigorous aspiration and vortexing may help. Ultimately, virus and protein aggregates may not be dispersed completely, and thus it is more important to collect the homogenous band from the gradient.

10. Alternately, 12-well plates can be used in which case ~2.5×10^5 Vero cells in 1 ml can be used. To keep consistent titers between formats, infection volumes should be adjusted accordingly and considered in the calculation of titers.
11. Usually, one in ten dilutions are used, although this can be further refined if more precision is required. Importantly, best results are obtained when tips are changed after each dilution step as the virus can stick to the tips and carry over to more diluted samples, leading to overestimations in titers.
12. This must be done fairly quickly since the agarose/DMEM mixture congeals rapidly as it cools. If several six-well plates need to be done simultaneously, it is best to do several small batches and make small quantities of agarose/DMEM mixtures each time.
13. After 24 h, viral plaques should be clearly visible by eye and can be counted directly. To facilitate the counting of plaques, we recommend fixing and staining Vero monolayers.
14. There exist several variants of the CT26 cell line. The CT26 cell line that can be obtained directly from ATCC is very resistant to VSVΔ51 infection. Some CT26 strains that have been subcloned, for example the CT26 clone 25 and CT26-lacZ strains, can be more sensitive to VSVΔ51.
15. Doses above 1×10^9 can lead to hind limb paralysis if the mice have not been previously immunized with the virus.
16. Typically, VSV replication can be observed at early time points (as early as 8–10 h) in the liver and spleen of mice, but this signal is transient. In contrast, replication in subcutaneous tumors increases over 24–72 h.
17. Do not drench tissues directly in ethanol as this may inactivate virus present at the surface.
18. For viral plaque assays starting from tissue homogenates, 2× DMEM media should be supplemented with penicillin–streptomycin in order to minimize the potential for contamination.
19. Up to 150 μl of sample can be collected at once using this method; however, serum can be easily obtained from as little as 20 μl.
20. Care should be taken not to collect any blood cells at this step.
21. Serum can be kept at 4°C for extended periods of time (easily up to a year).
22. It is important that the serum does not contain red blood cells as this leads to abnormally high viability values if a metabolic assay is used (e.g., spin down tubes first and take serum from the top).

23. The quantity of VSV used at this step depends on the expected range of antibody titer and the desired precision. For example, if low antibody titers are expected, small quantities of VSV should be used to increase precision. If a wide range of titers between samples is expected, more VSV is better as long as the quantity does not exceed the maximum antibody titer.

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